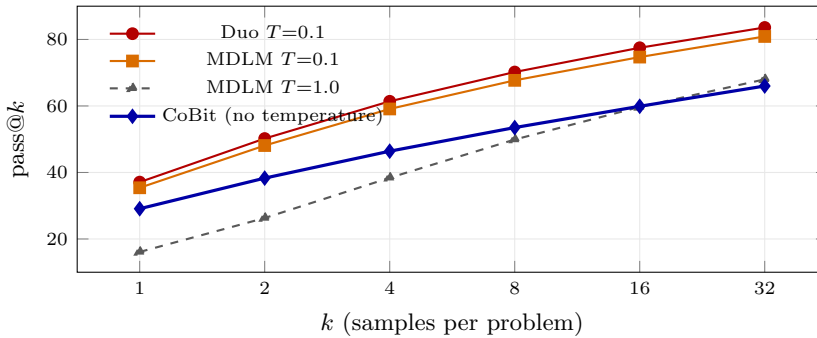


GSM8K pass@ k : first multi-sample comparison

csic47, 25 Aug 2026 — shard A = 256 random test problems (seed 0) — $K=32$ samples/problem, 1024 steps, fp32 — identical grader + identical problems for every method

The headline: our thesis is refuted. We predicted that low-temperature decoding buys discrete models pass@1 by destroying the sample diversity that pass@ k needs, so CoBit (no temperature) would dominate at $k > 1$. **It does not.** Duo at $T=0.1$ beats CoBit at *every* k and on majority vote. Lowering MDLM’s temperature raises pass@1 *and* pass@32 simultaneously.

	pass@1	pass@2	pass@4	pass@8	pass@16	pass@32	maj@32
CoBit (no temp.)	29.1%	38.3%	46.4%	53.5%	59.9%	66.0%	45.7%
Duo $T=0.1$	37.1%	50.2%	61.4%	70.2%	77.5%	83.6%	61.3%
MDLM $T=0.1$	35.4%	48.1%	59.1%	67.7%	74.7%	80.9%	59.4%
MDLM $T=1.0$	16.1%	26.3%	38.4%	49.9%	59.7%	68.0%	52.3%



What the numbers actually say

- 1. Low- T does not cost headroom; it adds it.** MDLM $T=1.0 \rightarrow 0.1$ lifts pass@1 16.1 $\rightarrow$ 35.4% *and* pass@32 68.0 $\rightarrow$ 80.9%. Sharpening improves the ceiling that post-training could realise, which is the opposite of the RLVR intuition (Yue et al., [arXiv:2504.13837](#)) we built the thesis on.
- 2. The trade is real but only in relative terms.** The $k=1 \rightarrow 32$ multiplier falls $4.2\times \rightarrow 2.3\times$ when MDLM is sharpened — diversity *is* being spent — but the absolute curve still rises everywhere. Relative climb is the wrong figure of merit.
- 3. CoBit’s own headroom is large but not competitive here.** 29.1 $\rightarrow$ 66.0% ($2.3\times$) with ESS=32/32, i.e. 32 genuinely independent churn samples, no tempering and no selection. It remains the best *continuous* model, and is best overall on Sudoku, but on GSM8K tuned discrete diffusion dominates at every sampling budget — not just at $k=1$, as the COLM draft concedes.

Status and caveats

Four of six cells are complete. duo.T1.0 crashed (OverflowError: int too large to convert to float in their metrics path, not OOM) and S-FLM has not run; both are needed to finish the temperature contrast. Numbers are on 256 of 1319 problems (± 3 points); the complementary 1063 are queued for CSD3 and merge by concatenation. pass@ k costs $k \times$ NFE, so these are *headroom* measurements, not matched-budget results. Reproduction note: our Duo $T=1$ single-sample is 19.5% vs. 17.2% published — 95% CIs overlap ([17.4, 21.6] vs. [15.2, 19.2]).

Implication for the workshop paper

The “diversity-vs-headroom” framing is dead and should not be written. The defensible paper is the *evaluation* one: no prior work (S-FLM included — one sample per problem, no pass@ k anywhere) reports multi-sample metrics on these benchmarks, and doing so overturns a natural intuition about what low-temperature decoding costs. CoBit is the continuous entrant, strongest on Sudoku; the GSM8K result is an honest negative that sharpens the existing concession.